UNIVERSITY OF ENERGY AND NATURAL RESOURCES
Department of Computer Science & Informatics
B.Sc. Computer Science Final Project

ACADEMIC Q&A & SECURE CONSULTATION PLATFORM

QCampus Connect

A Verified Academic Q&A Forum & Secure Consultation Platform for Higher Education

Institutional Case Study & Pilot Deployment: University of Energy and Natural Resources (UENR)

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COMP 450
FINAL RESEARCH PROJECT

🏛 Verified Academic Forum
Public departmental discussions & faculty Q&A exchange

🛡 Identity & Field Badges
Institutional credentials & specialization authority (CS, Crypto)

💬 Encrypted Consultation
Private 1-on-1 follow-up messaging (RSA-2048 + AES-256)

⛓ Immutable Blockchain Proof
Zero-gas Hyperledger Besu audit trail for non-repudiation

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Executive Overview of Presentation Structure (8 Core Topics)

PRESENTATION SCOPE: Structured 10-slide defense aligned with the Department of Computer Science & Informatics guidelines.

01

Background & Academic Reality in Higher Education

The Missing Academic Commons in Universities & The Failure of Informal Channels (Case Study: UENR)

🏛 The Missing Academic Commons in Higher Education

- **Static Portals vs. Interactive Commons:** Universities invest in grade portals and static LMS (lecture notes), but lack an open, interactive academic commons for student-faculty scholarly discussions.
- **Observed Reality at UENR:** To clarify coursework or project problems, students must physically hunt lecturers across campus buildings, queue outside offices, or make unscheduled phone calls.
- **Knowledge Fragmentation:** When a lecturer clarifies a concept to one student in private, that academic insight is lost to the remaining 100+ students in the cohort.

⚠️ Why Informal Channels (WhatsApp / Telegram) Fail

- **Cluttered & Ephemeral Streams:** Informal groups mix casual banter, memes, and administrative noise; serious academic inquiries and technical findings are quickly buried.
- **Redundant Faculty Burden:** Lecturers repeatedly answer the exact same questions for different students in private DMs with no searchable institutional knowledge archive.
- **QCampus Connect Solution:** A centralized university Q&A forum where discussions are public, searchable, and verified, with 1-on-1 messaging reserved only as a follow-up consultation tool.

📷 WORKFLOW COMPARISON & PROBLEM SCENARIO

Traditional Campus Wandering vs. QCampus Connect Digital Commons

- Left Side: Traditional Breakdown (Office queues, unscheduled phone calls, WhatsApp noise, duplicate DMs).
- Right Side: QCampus Connect Model (Public Forum Question -> Verified Faculty Answer -> Searchable Knowledge -> Optional 1-on-1 Consultation).
- Demonstrates how public Q&A solves the 1-to-many knowledge distribution barrier.
- Shows the seamless transition to client-encrypted direct messaging for deep private consultations.
- Eliminates late-night WhatsApp submission disputes with cryptographic timestamp proof.

CORE PARADIGM: Academic discussions must be open, centralized, and persistent so one lecturer explanation benefits the entire student cohort.

02

Problem Statement: Higher Education Communication Gap

Four Critical Institutional Bottlenecks in Campus Communication & Student-Faculty Collaboration

Problem 1: Academic Isolation & Inaccessibility

Office Queue Bottlenecks & Physical Barriers

- Students must physically pursue lecturers across campus buildings or wait in long queues outside offices.
- Off-campus, commuting, distance, or introverted students receive zero guidance outside limited physical office hours.
- Critical academic clarification is unavailable during revision and project deadlines.

Problem 2: Unqualified Advice as Expert Authority

Social Media Hearsay & Rumor Propagation

- In informal forums or WhatsApp groups, student upvotes and popularity outrank verified faculty expertise.
- Unverified rumors regarding exam dates, grading criteria, and coursework solutions circulate unchecked.
- No institutional badges exist to distinguish real professors from casual students.

Problem 3: Redundant Repetition & Faculty Burden

Knowledge Trapped in 1-on-1 Silos

- Lecturers spend dozens of hours answering identical conceptual questions repeatedly across separate private chats.
- Explanations given in 1-on-1 DMs are lost to the rest of the 100+ cohort.
- No searchable, persistent university knowledge repository exists to archive solved problems.

Problem 4: Ephemeral Records & Evidence Tampering

Lack of Cryptographic Non-Repudiation

- Social chat platforms allow 'Delete for Everyone', destroying evidence of critical interactions and assignment agreements.
- Database logs in standard LMS portals are mutable and vulnerable to timestamp falsification disputes.
- No independent, cryptographic audit trail exists to resolve submission disputes.

CORE RESEARCH QUESTION: "How can higher education institutions provide a structured, real-time academic discussion platform that eliminates physical accessibility barriers, verifies faculty authority, and preserves institutional knowledge? (Case Study: UENR)"

03

Aim & Specific Objectives (Case Study: UENR)

Research Scope, Mission Statement, and Five Primary Technical Deliverables

MAIN RESEARCH AIM

To design, implement, and benchmark QCampus Connect (QChat) — a verified academic Q&A and knowledge-sharing platform for higher education institutions (with UENR as the pilot case study) where students ask questions publicly, verified faculty provide authoritative guidance, and interactions carry cryptographic trust.

PRIMARY WORKSPACE

Obj 1: Real-Time Academic Q&A

Build a structured discussion forum categorized by Department (e.g., Computer Science) and Subject topics (e.g., Cryptography, OS, AI) with sub-50ms reactive WebSocket synchronization.

IDENTITY VERIFICATION

Obj 2: Authority & Field Badges

Cryptographically bind real institutional credentials (Student Index / Staff ID) and display verified departmental badges (CS) and specialization labels (Cryptography) on every faculty response.

SECURE SUB-FEATURE

Obj 3: 1-on-1 Encrypted DMs

Enable private student-faculty consultation initiated directly from Q&A threads using client-side RSA-OAEP 2048 & AES-GCM 256 encryption, paired with an educational BB84 QKD simulator.

IMMUTABLE TRUST

Obj 4: Blockchain Audit Trail

Anchor SHA-256 digests of answers and submissions to a permissioned Hyperledger Besu blockchain (QBFT, zero-gas smart contracts) for permanent non-repudiation and dispute resolution.

INSTITUTIONAL CONTROL

Obj 5: Admin & Governance Desk

Provide administrative tooling for departmental routing, lecturer credential approval, and cryptographic dispute audit verification via real student index/staff queries.

CENTRAL ARCHITECTURAL PRINCIPLE: "Q&A is the primary campus workspace. Direct messaging is a follow-up tool. Blockchain provides the proof."

04

Related Work & Platform Comparison

Comparative Analysis of Existing Higher Education Communication Channels vs. QCampus Connect

WhatsApp / Telegram *(Consumer Instant Messaging)*

• Strengths: Sub-second message delivery; transit encryption (Signal Protocol); ubiquitous adoption.

• Critical Academic Gaps: Unstructured, noisy chat streams; messages easily deleted ('Delete for Everyone'); zero institutional role verification; completely untrusted for formal academic audit.

Reddit / Stack Overflow *(Public Discussion Forums)*

• Strengths: Structured Q&A threading; community voting; topic tags.

• Critical Academic Gaps: Completely detached from university curricula; pseudonymous accounts; upvote karma rewards popularity over academic expertise; posts can be silently edited with no proof.

Canvas / Moodle LMS *(Coursework Management (VLE))*

• Strengths: Assignment submission modules; gradebook tracking; course file hosting.

• Critical Academic Gaps: Static file repositories; lacks real-time interactive peer-faculty discussions; relies on mutable SQL database logs without cryptographic non-repudiation.

Slack / Microsoft Teams *(Enterprise Team Collaboration)*

• Strengths: Organized departmental channels; file sharing; role-based access control.

• Critical Academic Gaps: Expensive licensing; centralized server custody (admins can purge or edit records); no independent client-side encryption or blockchain proof.

RESEARCH SYNTHESIS: QCampus Connect uniquely unifies: (1) Sub-50ms reactive WebSocket discussion (Convex), (2) Institutional identity & specialization badges, (3) Client-encrypted 1-on-1 consultation (RSA/AES), and (4) Zero-gas immutable blockchain provenance (Hyperledger Besu).

05

Methodology & System Architecture

Three-Tier Enterprise Architecture: Client-Side WebCrypto, Real-Time Serverless & Blockchain Layer

1. PRESENTATION LAYER (Browser — React 19 SPA)

Tech Stack: React 19 + TypeScript + Vite + Tailwind CSS + Web Crypto API

• Core Modules: Verified Q&A Forum (Primary), Direct Messages (Sub-feature), Admin Audit Desk, BB84 Simulator.

• Security: RSA-OAEP 2048 keypairs & AES-GCM 256 generated locally in browser; private keys stored securely in IndexedDB.

2. APPLICATION LAYER (Convex Serverless Platform)

Tech Stack: Convex Serverless Cloud & WebSocket Engine (~50ms Reactive Sync)

• Document Store: users, departments, questions, answers, chatRooms, encryptedFiles.

• Functions: getQuestions, askQuestion, addAnswer, getOrCreateRoom, verifyUser.

• Unread count tracking and live UI state synchronization.

3. VERIFICATION LAYER (Hyperledger Besu + Solidity)

Tech Stack: Hyperledger Besu Private Ethereum Network (QBFT Consensus, ChainId 1337, Zero Gas)

• Smart Contract: MessageVerifier.sol (recordHash, verifyUserWithIndex, verifyUserWithStaffId).

• Audit Trail: Asynchronous SHA-256 digest anchoring (~0.3-0.5s block write) for non-repudiation.

THREE-TIER SYSTEM ARCHITECTURE & CRYPTO FLOW

End-to-End Reactive WebSockets & Asynchronous Besu Blockchain Anchoring

• User Browser: Web Crypto API generates RSA-2048 & AES-256 keys locally. Private key saved to IndexedDB.

• Convex Serverless: Real-time WebSockets sync questions and answers to all active clients in <50ms.

• Besu Blockchain: Asynchronous background worker commits SHA-256 hash digests to MessageVerifier.sol.

• Admin Audit Desk: Independent verifier queries smart contract to prove authenticity without trusting server.

• Zero Gas Fees: QBFT consensus on private Besu network eliminates all transaction costs for university.

ARCHITECTURAL INSIGHT: Users experience ~50ms instant messaging via Convex WebSockets, while blockchain audit logging occurs asynchronously in the background (~0.3-0.5s) without blocking the UI or charging gas fees.

06

Academic Q&A & Real-World Identity Workflow

End-to-End Six-Step Cryptographic Verification & Consultation Lifecycle (From Forum Question to Audit Proof)

🔄 Six-Step End-to-End Verification Lifecycle

- **Step 1: Question Posting & Department Classification:** Student posts question categorized by Department (e.g. Computer Science) and Topic tag (e.g. Cryptography). Thread is indexed publicly for all peers to study.
- **Step 2: Faculty Answer & Cryptographic Badge Binding:** Verified faculty member answers; system automatically validates credentials and renders badges: 🏛 Computer Science & 📚 Cryptography, proving expert authority.
- **Step 3: 1-on-1 Private Consultation Trigger:** If deeper private assistance is required, the student clicks '💬 Message' directly on the faculty's answer to launch an end-to-end encrypted direct chat session.
- **Step 4: Asynchronous Blockchain Hash Anchoring:** System computes SHA-256 digest of question, answer, or assignment submission and registers it on Hyperledger Besu (MessageVerifier.sol) with block timestamp.
- **Step 5: Admin Identifier Search (Audit Desk):** Admin or examiner searches student by real institutional credential (Index: UEB3509022 / Email: osei@uenr.edu.gh / Staff: PS001) in the Admin Audit Desk.
- **Step 6: Certified Verdict & Non-Repudiation Report:** System compares live database digest against immutable Besu block timestamp, displaying instant verdict: '✅ Authentic Submission' with permanent tx hash.

📷 ADMIN AUDIT DESK & VERIFICATION PROOF MODAL

Live Blockchain Timestamp Verification & Tamper Detection Interface

- Student Identifier Search: Query index number 'UEB3509022' or staff ID 'PS001'.
- Submission & Post Inspector: Shows submitted coursework, file attachments, and on-chain tx hashes.
- Cryptographic Audit Trail Modal: Compares database record vs. immutable Besu block record.
- Certified Verdict Banner: Displays '✅ Authentic Submission' confirmation.
- Tamper Detection Guarantee: If database record is altered, SHA-256 hash mismatch alerts admin instantly.

AUDIT GUARANTEE: Even if a malicious actor alters a database submission timestamp or edits text directly in SQL, the SHA-256 block hash mismatch on Hyperledger Besu immediately exposes the tampering.

07

Empirical Results & Live System Demo

Empirical Test Suite (100% PASS), System Latency Benchmarks, and Live Feature Demonstration

📊 Empirical Verification Suite & Benchmarks (100% PASS)

- **Production Build:** Vite compiled 244 modules with 0 TypeScript errors in 3.36s (dist: 780 kB js, 94 kB css).
- **RSA-OAEP 2048 Keygen:** Generated in ~140ms in browser; private key committed to IndexedDB (✅ PASS).
- **AES-GCM-256 Cipher:** Payload encryption executed in ~2ms; invalid decryption key throws error (✅ PASS).
- **BB84 QKD Simulator:** 256-bit sifted key derived in ~12ms; aborts correctly if QBER > 11% (✅ PASS).
- **Besu Smart Contract:** recordHash logs SHA-256 digest; duplicate write reverts; role checks enforced (✅ PASS).
- **Convex Reactive Latency:** WebSocket real-time delivery averaged ~50ms (well under the 200ms target).
- **Besu Block Anchoring:** Asynchronous transaction confirmation in ~0.3–0.5s with zero gas fees.

📷 LIVE SYSTEM INTERFACE: Q&A FORUM & DIRECT MESSAGING

Panel-Ready Demonstration: Verified Discussions & Private Consultation Flow

- Verified Q&A Forum Feed: Questions tagged by Department ('Computer Science') and Topic ('Cryptography').
- Lecturer Answer View: Verified identity badge (🏛 CS, 📚 Cryptography) establishing expert authority.
- Direct Consultation Launch: '💬 Message' button on lecturer's answer seamlessly opens encrypted DM.
- BB84 Simulation Interface: Live visualization of quantum state transmission, sifting, and error rate check.
- Verified Student Index: Osei Nana Kwaku (UEB3509022) submission generating live Besu blockchain receipt.

DEMO VERDICT: The live demonstration proves that high-speed reactive discussions (<50ms) and permanent blockchain non-repudiation can coexist in a production-ready, zero-error web application.

Limitations, Future Work & Conclusion

System Boundaries, Recommended Technical Roadmap, and Final Research Defense Conclusions

1. PROGRAM-SPECIFIC Q&A AND UPDATES

The platform will be narrowed from a broad departmental structure to program-specific communities, allowing students and lecturers within the same program to access relevant academic questions, discussions, news, and updates without unnecessary information from other programs.

2. COURSE-SPECIFIC CHATROOMS

Each course will have a dedicated chatroom where the lecturer and enrolled students can interact directly about course-related topics. This creates a focused space for asking questions, clarifying concepts, sharing resources, and discussing course activities.

3. AUTOMATIC IDENTITY VERIFICATION

An AI-assisted identity verification system will compare information provided on QChatCampus with authorized university records, such as student identity details, name, and department/program. The goal is to automatically verify users and complete the verification process within seven days or less.

4. DEPARTMENTAL NEWS & NOTICE BOARD

A dedicated departmental news and notice section will allow lecturers to publish announcements, updates, images, and files. Students and lecturers within the department can search posts by title, hashtags, department, or recent updates, while also commenting and interacting with published notices.

FINAL TAKEAWAY: "Fast like social chat. Structured like an academic forum. Trustworthy like a signed paper trail." | Supervisor: Dr. Peter Nimbe (PS001) | Dept. of Computer Science & Informatics | UENR

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Limitations, Future Work & Conclusion

System Boundaries, Recommended Technical Roadmap, and Final Research Defense Conclusions

⚠️ System Constraints

Current Boundaries

• Single-Node Blockchain: Besu runs in local container; requires multi-node consortium for distributed BFT.

• Browser Storage Limits: IndexedDB quotas (50-250MB) may bottleneck large ledger caches.

• Standalone Authentication: Independent user directory; lacks university SAML/SSO link.

• Web-Only Client: Desktop optimized; lacks native mobile application.

✅ Key Research Conclusions

Defense Summary

• Academic Commons Established: Centralized Q&A forum eliminates office queues and private chat silos.

• Verified Scholarly Authority: Cryptographic badges eradicate fake announcements and rumors.

• Submission Disputes Solved: SHA-256 block timestamps replace easily falsified screenshots.

• Production-Ready Blueprint: 0-error build ready for departmental adoption.

FINAL TAKEAWAY: "Fast like social chat. Structured like an academic forum. Trustworthy like a signed paper trail." | Supervisor: Dr. Peter Nimbe (PS001) | Dept. of Computer Science & Informatics | UENR